0.1 Backdoor

Claim 1 *If* $OP =$

$$
\begin{tikzcd}
\underline{z} \arrow[d] \arrow[dr] & \\
\underline{x} \arrow[r] & \underline{y}
\end{tikzcd}
$$

then

$$
\begin{aligned}
P(y|\mathcal{D}\underline{x} = x) &= \sum_z P(y|x, z)P(z) && (1) \\
&= \begin{tikzcd}
\mathbb{E}z \arrow[dr] & \\
x \arrow[r] & y
\end{tikzcd} && (2)
\end{aligned}
$$

proof:

See `https://github.com/rrtucci/dag_iden_detector` for computer generated numerical proof.

QED

0.2 Frontdoor

Claim 2 *If* $OP =$

$$
\begin{tikzcd}
& \underline{c} \arrow[dl] \arrow[dr] & \\
\underline{x} \arrow[r] & \underline{m} \arrow[r] & \underline{y}
\end{tikzcd}
$$

then

$$
\begin{aligned}
P(y|\mathcal{D}\underline{x} = x) &= \sum_m \left[\sum_{x'} P(y|m, x')P(x') \right] P(m|x) && (3) \\
&= \begin{tikzcd}
& \mathbb{E}x' \arrow[dr] & \\
x \arrow[r] & \sum m \arrow[r] & y
\end{tikzcd} && (4)
\end{aligned}
$$

proof:

See `https://github.com/rrtucci/dag_iden_detector` for computer generated numerical proof.

QED

0.3 Napkin1

Claim 3 *If* $OP =$
$$\begin{tikzcd} & & \underline{u}_1 \arrow[dll] \arrow[drrr] & & \\ & & \underline{u}_2 \arrow[dll] \arrow[dr] & & \\ \underline{w} \arrow[rr] & & \underline{z} \arrow[r] & \underline{x} \arrow[r] & \underline{y} \end{tikzcd}$$
then

$$\begin{aligned} P(y \mid \mathcal{D}\underline{x} = x) &= \frac{\sum_{w,z} P(x, y|w, z)P(w)P(z)}{\sum_y \ \ num} &(5)\\ &= \frac{\sum_{w,z} P(y|x, w, z)P(x|w, z)P(z)P(w)}{\sum_y \ \ num} &(6)\\ &= \frac{1}{\sum_y \ \ num} \begin{tikzcd} \mathbb{E}w \arrow[rrr, bend left] \arrow[rr, bend right] & \mathbb{E}z \arrow[r] \arrow[rr, bend left] & x \arrow[r] & y \end{tikzcd} &(7) \end{aligned}$$

proof:

See `https://github.com/rrtucci/dag_iden_detector` for computer generated numerical proof.
QED

0.4 Napkin2

Claim 4 *If* $OP =$
$$\begin{tikzcd} & & \underline{u}_1 \arrow[dll] \arrow[drrr] & & \\ & & \underline{u}_2 \arrow[dll] \arrow[dr] & & \\ \underline{w} \arrow[rr] & & \underline{z} \arrow[r] & \underline{x} \arrow[r] & \underline{y} \end{tikzcd}$$
then

$$\begin{aligned} P(y \mid \mathcal{D}\underline{x} = x) &= \sum_z P(y|x, z)P(z) &(8)\\ &= \begin{tikzcd} & \mathbb{E}z \arrow[dr] & \\ x \arrow[rr] & & y \end{tikzcd} &(9) \end{aligned}$$

proof:

This claim is INCORRECT. See `https://github.com/rrtucci/dag_iden_detector` for computer generated numerical counterexample.
QED

0.5 Napkin3

Claim 5 *If* $OP =$
$$\begin{tikzcd} & \underline{u}_1 \arrow[ddl] \arrow[ddrr] & & \\ & \underline{u}_2 \arrow[dl] \arrow[dr] & & \\ \underline{w} \arrow[r] & \underline{z} \arrow[r] & \underline{x} \arrow[r] & \underline{y} \end{tikzcd}$$
then

$$P(y \mid \mathcal{D}\underline{x} = x) = \sum_w P(y|x,w)P(w) \tag{10}$$

$$= \begin{tikzcd} & \mathbb{E}w \arrow[d] \\ x \arrow[r] & y \end{tikzcd} \tag{11}$$

proof:

This claim is INCORRECT. See `https://github.com/rrtucci/dag_iden_detector` for computer generated numerical counterexample.

QED

0.6 Napkin4

Claim 6 *If* $OP =$
$$\begin{tikzcd} & \underline{u}_1 \arrow[ddl] \arrow[ddrr] & & \\ & \underline{u}_2 \arrow[dl] \arrow[dr] & & \\ \underline{w} \arrow[r] & \underline{z} \arrow[r] & \underline{x} \arrow[r] & \underline{y} \end{tikzcd}$$
then

$$P(y \mid \mathcal{D}\underline{x} = x) = \frac{\sum_w P(x,y|w,z)P(w)}{\sum_y \ num} \tag{12}$$

$$= \frac{\sum_w P(y|x,w,z)P(x|w,z)P(w)}{\sum_y \ num} \tag{13}$$

$$= \frac{1}{\sum_y \ num} \begin{tikzcd} \mathbb{E}w \arrow[rrr, bend left] \arrow[rr, bend right] & z \arrow[r] \arrow[rr, bend left] & x \arrow[r] & y \end{tikzcd} \tag{14}$$

proof:

This claim is INCORRECT. See `https://github.com/rrtucci/dag_iden_detector` for computer generated numerical counterexample.

QED

0.7 Napkin5

Claim 7 *If* $OP =$
$$
\begin{tikzcd}
 & \underline{u}_1 \arrow[ddl] \arrow[ddrr] & & \\
 & \underline{u}_2 \arrow[dl] \arrow[dr] & & \\
\underline{w} \arrow[r] & \underline{z} \arrow[r] & \underline{x} \arrow[r] & \underline{y}
\end{tikzcd}
$$
then

$$
P(y \mid \mathcal{D}\underline{x} = x) = \frac{\sum_{w_1, w_2} P(y|x, w_1) P(x|w_2) P(w_1) P(w_2)}{\sum_y \ num} \tag{15}
$$

$$
= \frac{1}{\sum_y \ num} \quad
\begin{tikzcd}
\mathbb{E}w_1 \arrow[ddrr] & & \\
\mathbb{E}w_2 \arrow[dr] & & \\
 & \underline{x} \arrow[r] & \underline{y}
\end{tikzcd} \tag{16}
$$

proof:

This claim is INCORRECT. See `https://github.com/rrtucci/not-do-calculus` for computer generated numerical counterexample.

QED